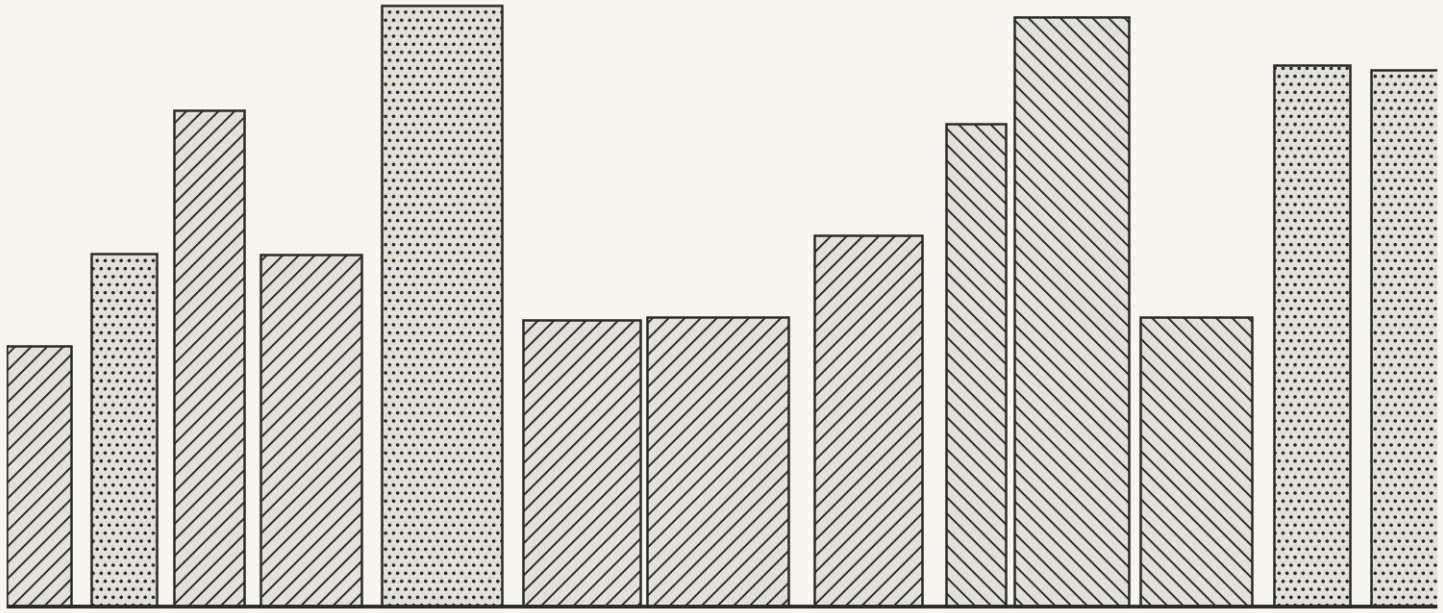
SORTING BY DOMINANCE

— a short tour of NSGA-II —



Your Name

Your Institution

Month Year

PART I

THE PROBLEM

1.1 Two goals that disagree

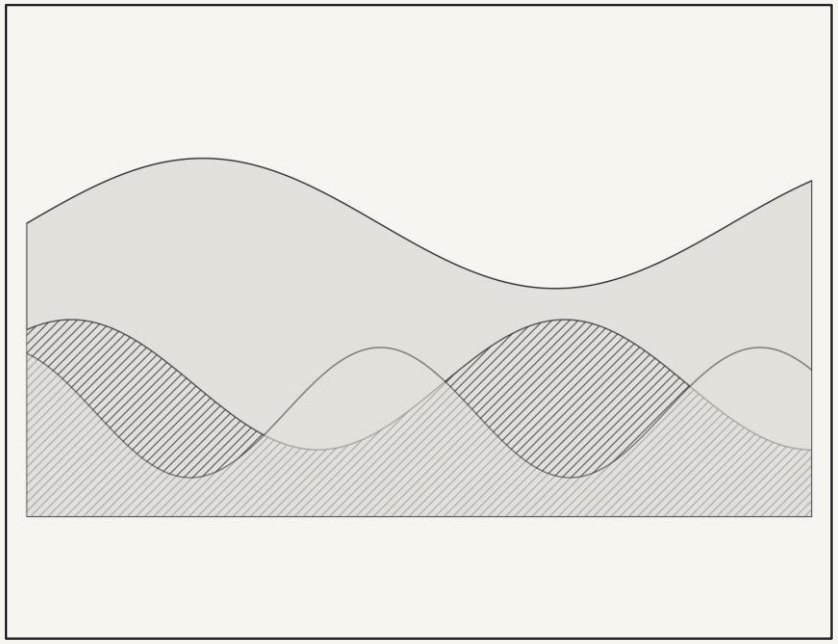
1.2 Dominance

1.3 The Pareto front

TWO GOALS THAT DISAGREE

— no single best answer —

TRADE-OFF



— placeholder illustration —

A cheaper design is usually a weaker one. When two goals pull in opposite directions, one solution cannot be best at both.

Multiobjective optimisation keeps the whole set of fair compromises instead of forcing one weighted score.

DOMINANCE

— the only comparison NSGA-II needs —

$$x \prec y \iff f_m(x) \leq f_m(y)\ \forall m \text{ and } f_k(x) < f_k(y) \text{ for some } k$$

— x dominates y when it is no worse everywhere and better somewhere (all goals minimised) —

Solutions that nothing dominates form the Pareto front.

PART II

THE ALGORITHM

2.1 The loop

2.2 Three ingredients

2.3 A worked table

ONE GENERATION

— parents and children compete together —

I. Pick parents by binary tournament.

II. Make children with crossover and mutation.

III. Merge parents and children into one pool.

IV. Sort the pool into fronts by dominance.

V. Fill the next population front by front.

VI. Break the tie in the last front by crowding distance.

THREE INGREDIENTS

— what makes it NSGA-II —

I · NON-DOMINATED SORTING

Rank every solution by the front it belongs to. Front one is the best.

II · CROWDING DISTANCE

Prefer solutions in sparse regions so the front stays spread out.

III · ELITISM

Parents compete with children, so a good solution is never lost.

Reference: Deb, Pratap, Agarwal and Meyarivan (2002), IEEE Transactions on Evolutionary Computation 6(2).

CROWDING DISTANCE

— how isolated is a solution —

$$d_i = \sum_{m=1}^{M} \frac{f_m^{(i+1)} - f_m^{(i-1)}}{f_m^{\max} - f_m^{\min}}$$

— neighbours are taken after sorting the front by goal m; the two ends get infinite distance —

A larger value means fewer neighbours nearby, so the solution is kept.

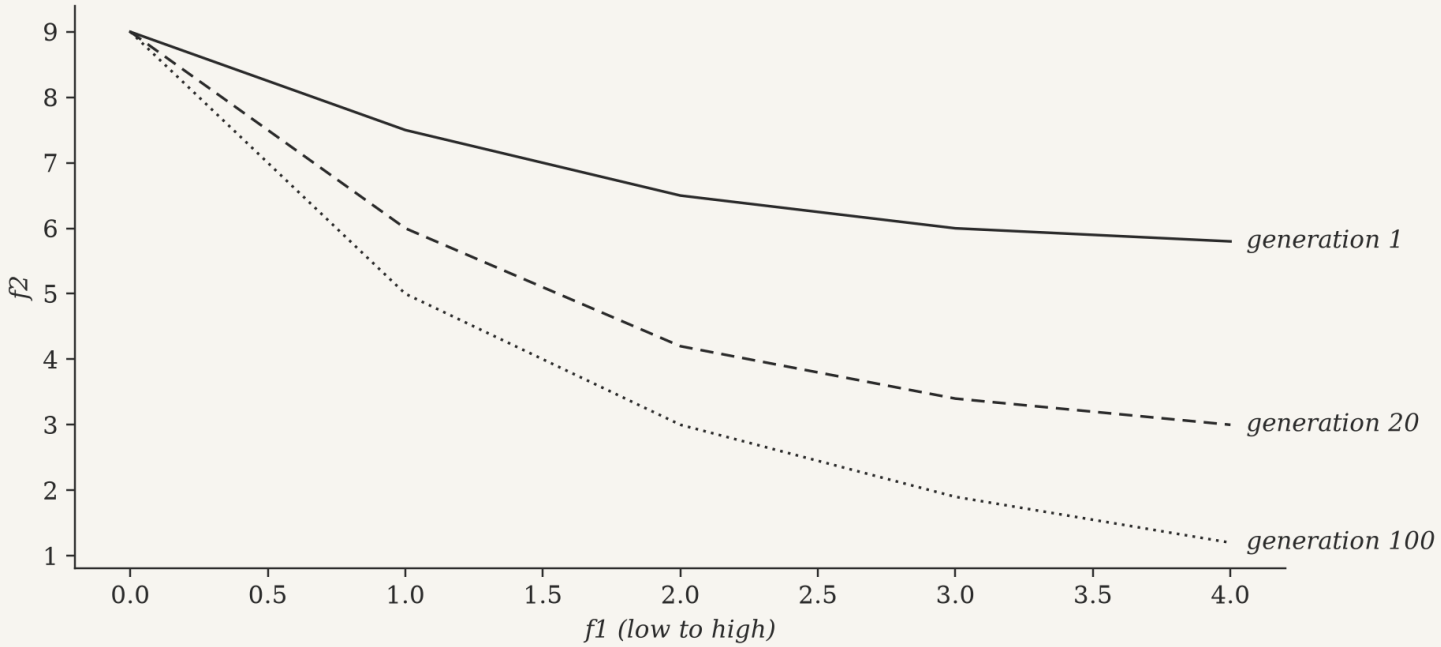
A WORKED TABLE

— four invented solutions, two goals to minimise —

SOLUTION	F1	F2	FRONT	CROWDING
A	1	9	1	*inf*
B	3	5	1	*2.00*
C	4	6	2	*inf*
D	7	2	1	*inf*

A FRONT CLOSING IN

— a sketch with invented values —

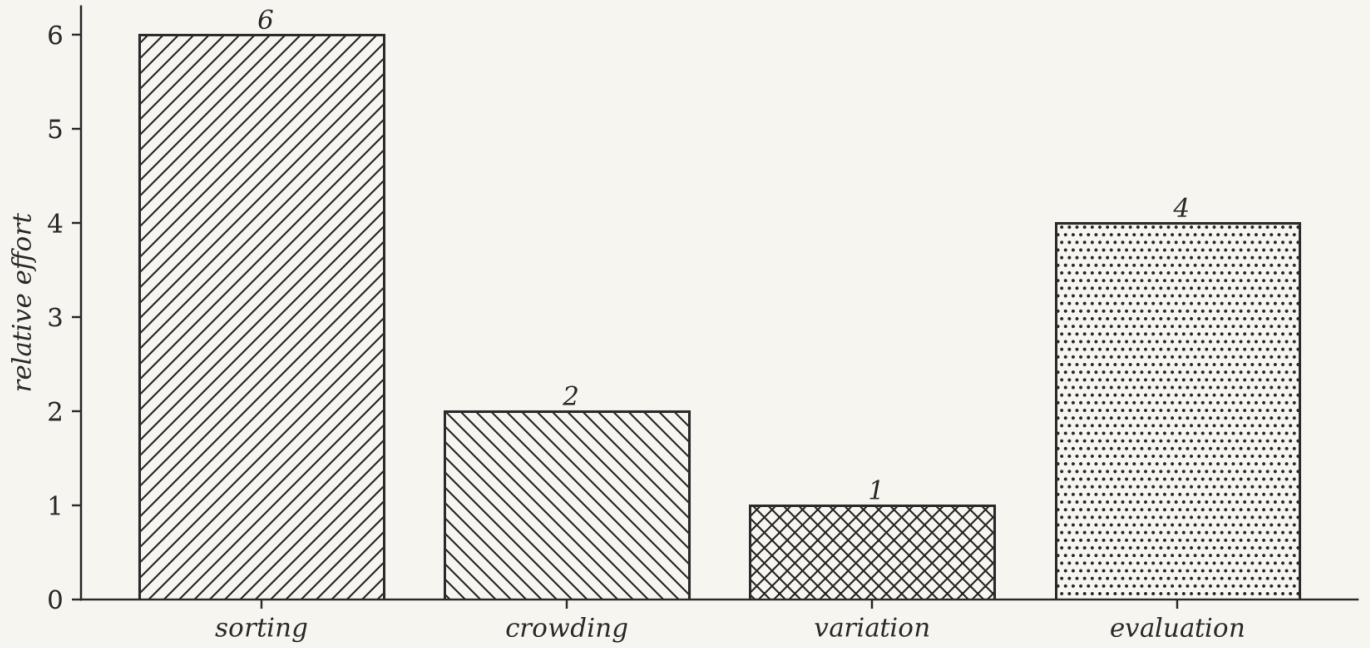


— invented values, only to show the idea —

With more generations the set of compromises moves toward the lower left and spreads along it.

WHERE THE COST GOES

— sorting dominates the work —

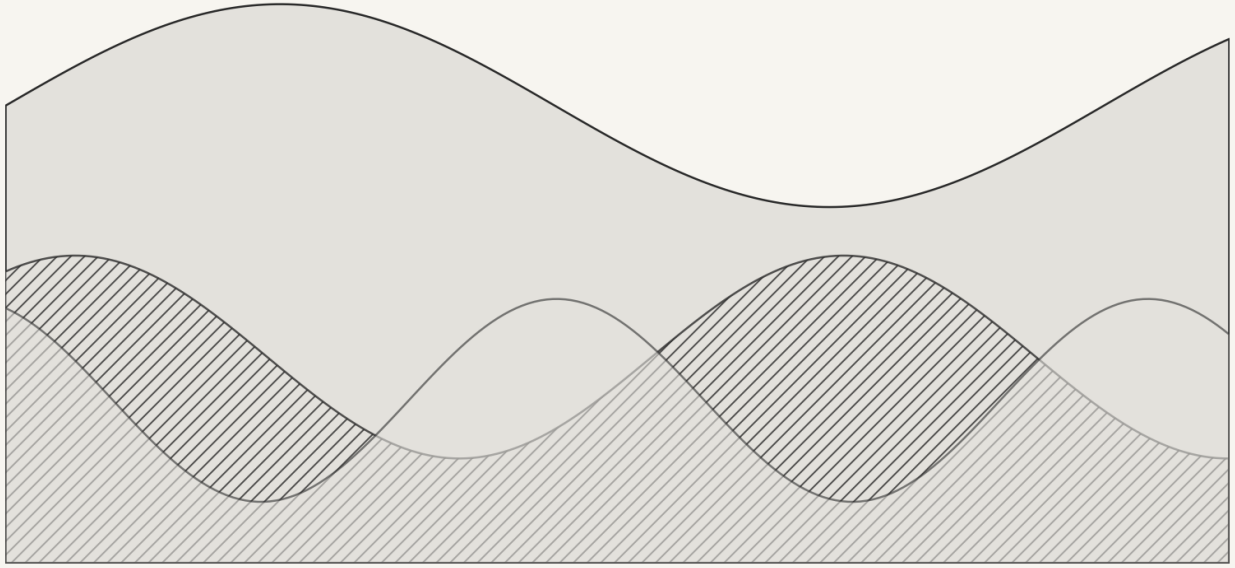


— invented values, evaluation cost depends on the problem —

Non-dominated sorting costs on the order of M times N squared comparisons per generation (Deb et al., 2002).

THANK YOU

— questions and corrections welcome —



Dominance orders the solutions, crowding keeps them spread out, and elitism keeps what works.

— Your Name —